

Sandesh Bhandari

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Research Interests

Statistical and machine learning approaches to urban spatial data: how cities can be modeled, predicted, and understood through geographic and earth-systems data at fine spatial scale. My focus is on problems where the spatial relationships between observations are fundamental to how the data should be modeled, explored, and interpreted.

Skills

GIS & Spatial Analysis

ArcGIS Pro, QGIS, GeoPandas, Rasterio, Shapely, GDAL/OGR, OSMnx, PySAL, NetworkX, Google Earth Engine

Programming & Data

Python, JavaScript, TypeScript, SQL, Bash, NumPy, SciPy, Pandas, Polars

AI & Machine Learning

PyTorch, PyTorch Lightning, scikit-learn, LightGBM, SHAP, Hugging Face Transformers, semantic segmentation (SegFormer), OpenCV

Visualization, Web & Graphics

D3.js, THREE.js, GLSL/WebGL, Canvas 2D, Node.js, Tauri, Leaflet, Folium, Matplotlib, Seaborn, Figma, Illustrator, Photoshop, InDesign

Education

University of Toronto, Faculty of Arts and Science

2026 – 2028

Master of Science in Planning (MScPl), Department of Geography and Planning

University of Toronto, Faculty of Arts and Science

2026

HBSc. in Geography, Earth and Data Sciences, with High Distinction

- GPA: 3.92/4.0, Dean's List Scholar
- Nicholas Wemyss Undergraduate Explorers Fund Award
- John Horner Undergraduate Scholarship in Geography
- University of Toronto Research Excellence Award (UTEA)
- Undergraduate Computer Applications Award

Research & Professional Experience

University of Toronto, Spatial Analysis of Urban Systems Lab

May 2026 – Present

Research Assistant, street-level heat exposure study (supervisor: Ignacio Tiznado-Aitken)

- Lead a study that maps street-level heat risk along Toronto's walking, cycling and transit network from satellite surface temperature and street-view imagery; first author of the manuscript in preparation.
- Built the network as 347,298 points at 30 m spacing, including 9,054 transit stops, labelled heat risk from a multi-year Landsat 8/9 summer composite in Google Earth Engine, and thinned the sample to 70,161 points to limit spatial leakage.
- Extracted sky view, tree canopy and built-surface features by semantic segmentation of street-level imagery, and trained a LightGBM classifier under spatially blocked cross-validation, interpreted with SHAP and GeoShapley.

University of Toronto, Department of Geography & Planning

January 2026 – Present

Research Assistant, Major Streets study (supervisors: Paul Hess and Michael Piper)

- Conduct spatial and parcel-level geometry analysis of the residential lots along Toronto's major streets, to assess where the City's policy allowing small apartment buildings on these streets can add housing.
- Built a reproducible Python (GeoPandas) workflow that identifies each lot's street frontage, measures its width, depth and area from the parcel geometry, and joins zoning, land use and building age from open City data.
- Validated the measurements against an independent dataset, reproduced the City's count of lots with development potential, and ran a sensitivity analysis on the lot-size thresholds behind it.

City of Toronto, Planning Research and Analytics
Research Trainee

May 2026 – August 2026

- Surveyed Toronto businesses in person to gather employment and land-use data that inform municipal planning and future city services.
- Verified site records and maintained the statistical databases that support project reporting.

Web Developer

2019 – 2023

Independent Contractor

- Built and deployed responsive, custom web applications (React, Node.js, MongoDB) for 15+ local small businesses.
- Managed end-to-end project lifecycles from scoping and UI/UX design to production deployment, hosting, and client handoff.

Publications & Presentations

Bhandari, S., & Tiznado-Aitken, I. (in preparation). Micro-scale thermal exposure along active-transport networks: A machine-learning approach using street view imagery.

Bhandari, S. (2025). *Re-Zoning the Yard*. ArcGIS StoryMap, featured in the [Virtual Map Gallery](#) of Esri Canada's GIS in Education and Research Conference.

Bhandari, S. (2026). *Kensington Market Heritage Conservation District Plan: A Critical Analysis*. Research presentation. [Slides](#)

Teaching Experience

University of Toronto, Geography & Planning

October 2026 – Present

Teaching Assistant and Research Assistant, GGR375 Introduction to Programming in GIS

- Teaching Assistant (Winter 2027): support third-year students as they learn to process and analyse geospatial data in Python (GeoPandas, Rasterio, PySAL) on the university JupyterHub.
- Research Assistant, Computation & Data Science Education Faculty Fellowship: help redesign GGR375 to add AI-assisted coding and critical evaluation of AI output, keeping its core GIS programming foundations: reproducible Jupyter notebooks, new exercises and assignments.
- Co-run the student feedback phase: survey, interviews and focus groups with former GGR375 students on how they used AI and on the course's "Responsible Use of AI" statements.

Selected Projects

Full portfolio: <https://snes19xx.github.io>

Spatial and Socioeconomic Distribution of Toronto's Urban Heat Islands [\[Link\]](#)

Landsat 8/9 heat, NDVI and census data for Toronto, with Getis-Ord Gi* and Moran's I hotspot analysis.

Atlas of Canadian Wildfires [\[Link\]](#)

Interactive D3 atlas of 20,946 Canadian fires (1959–2025) with kernel density, hotspot and trend layers.

Rebuilding the 510 Spadina from its own data [\[Link\]](#)

Monte Carlo streetcar simulation from TTC GTFS data, showing 39% faster median trips and 67% better headway reliability.

three.lab [\[Link\]](#)

Browser-based three.js tool for editing 3D models, building OpenStreetMap city models, and exporting GLB, SVG and pen-plotter output.

Professional & Community Affiliations

Kensington Market Community Land Trust (KMCLT), *Community Member*

Canadian Institute of Planners (CIP), *Student Member since 2023*

American Planning Association (APA), *Student Member since 2023*